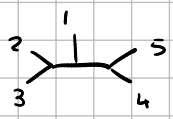
Generic amplitude: arbitrary linear combination of all ordered Feynman diagrams for n external legs.

D-subsets

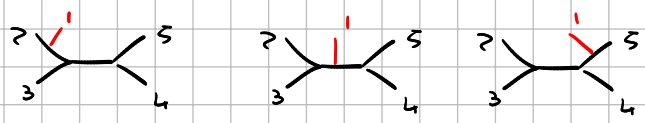
A D-subset at n-point is the subset of the generic amplitude that contains all diagrams obtained by starting with a particular Feynman diagram, removing leg 1, then reattaching it in all possible ways (within the order).

For example: $n=5$, start with

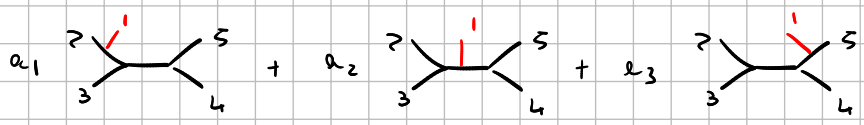


- remove leg 1: $\rightarrow$

- add back leg 1 in all ways:

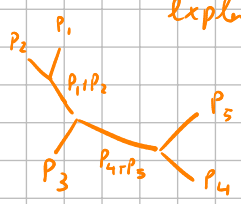


the associated D-subset is



$$= \frac{a_1}{X_{13}\,X_{41}} + \frac{a_2}{X_{24}\,X_{41}} + \frac{a_3}{X_{24}\,X_{52}}$$

explanation

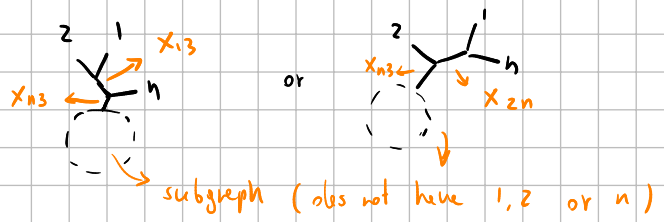


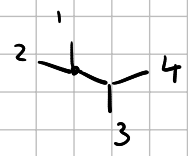
$$= \frac{1}{(p_1+p_2)^2}\,\frac{1}{(p_4+p_5)^2} = \frac{1}{X_{13}}\,\frac{1}{X_{41}}$$

Def: A "boundary propagator" is an X_{ij} that depends on z under a BCFW shift (so $X_{13}, \dots, X_{1n-1}$ and $X_{24}, \dots, X_{2n}$)

Now, if B is our generic amplitude at n points, we split it in groups depending on the number of boundary propagators each term has. This is the same as splitting $B = B_m + B_{m-1} + \dots$ from earlier, since a term with k boundary propagators scales as z^{-k} when $z \to \infty$.

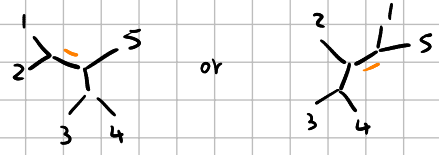
• A graph with 1 boundary propagator looks like





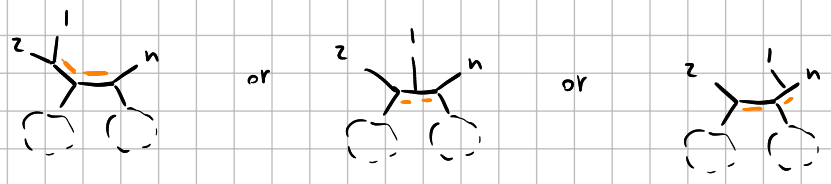
$= \frac{1}{x_{12}}$ $n = 4$ (only 1)

$n = 5$

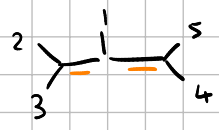


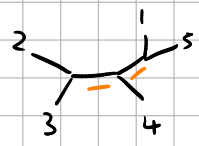
$= \frac{1}{x_{13}\, x_{35}}$ $= \frac{1}{x_{25}\, x_{35}}$

• 2 boundary operators



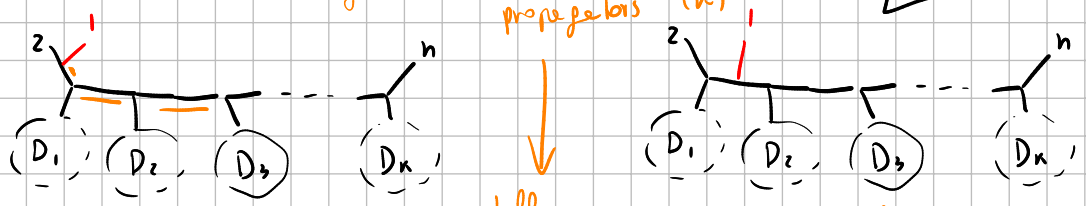
$n = 5$ 2 boundary



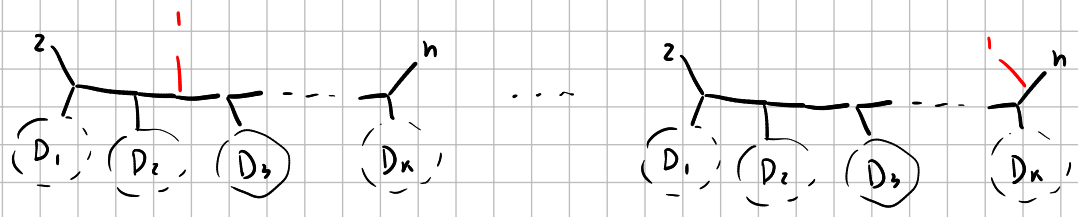


• k boundary operators:

orange lines count # boundary propagators (k)



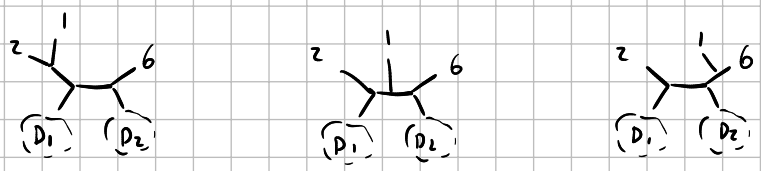
tells us these graphs scale as z^{-k}



It should be clear from the image that removing leg 1 and re-adding it does not change the # of boundary propagators, so we can partition each B_{-k} in D-subsets (by going through all choices of subgraphs $D_1, \dots D_k$).

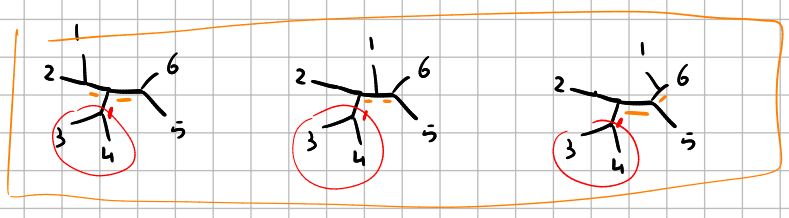
Let's see an example: $n = 6$, 2 boundary propagators.

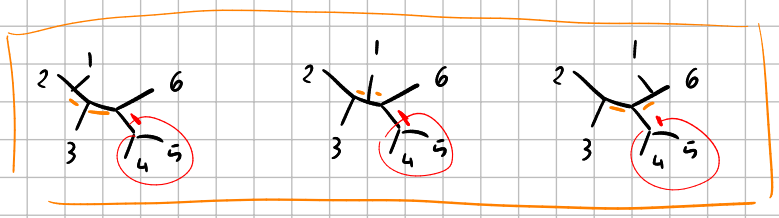
must look like



the choices for D_1, D_2 are: $\left(D_1 = \begin{smallmatrix} & 1 & \\ 3 & & \\ & & 4 \end{smallmatrix} \quad D_2 = \begin{smallmatrix} 1 \\ 5 \end{smallmatrix} \right)$ and $\left(D_1 = \begin{smallmatrix} 1 \\ 3 \end{smallmatrix} \quad D_2 = \begin{smallmatrix} & 1 & \\ 4 & & 5 \end{smallmatrix} \right)$

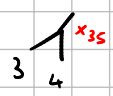
so there are 2 D-subsets:

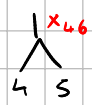
$S_1 =$ 

$S_2 =$ 

Since $S_1 + S_2 = B_{-2}$ we know it must satisfy 1-zero if B does.

However, so do S_1 and S_2 independently, since

S_1 has a factor of X_{35} coming from the  piece

and S_2 has a factor of X_{46} coming from the  piece.

$\rightarrow \quad S_1 + S_2 = \frac{1}{X_{35}}(\cdots) + \frac{1}{X_{45}}(\cdots) = 0 \quad \forall\, X_{35}, X_{45}$ only if

the coefficients of $\frac{1}{X_{35}}$ and $\frac{1}{X_{45}}$ vanish independently $\Rightarrow S_1 = 0,\ S_2 = 0$.

Moreover, imposing the 1-zero uniquely determines S_1 and S_2 up to an overall coefficient. Note that

$$S_1 = \frac{1}{X_{35}}\left(\frac{a_1}{X_{13}\, X_{15}} + \frac{a_2}{X_{25}\, X_{15}} + \frac{a_3}{X_{25}\, X_{26}} \right) = 0$$

$$\Leftrightarrow \quad \frac{a_1}{X_{13}\, X_{15}} + \frac{a_2}{X_{25}\, X_{15}} + \frac{a_3}{X_{25}\, X_{26}} = 0$$

using the fact that the 1-zero constrains $X_{26} = -X_{13}$

we get $$0 = \frac{a_1}{X_{13}\, X_{15}} + \frac{a_2}{X_{25}\, X_{15}} - \frac{a_3}{X_{25}\, X_{13}}$$

$$= \frac{a_1 X_{25} + a_2 X_{13} - a_3 X_{15}}{X_{13} X_{15} X_{25}} = 0$$

so it must be $\quad a_1 X_{25} + a_2 X_{13} - a_3 X_{15} = 0$

$X_{25} = X_{15} - X_{13}$

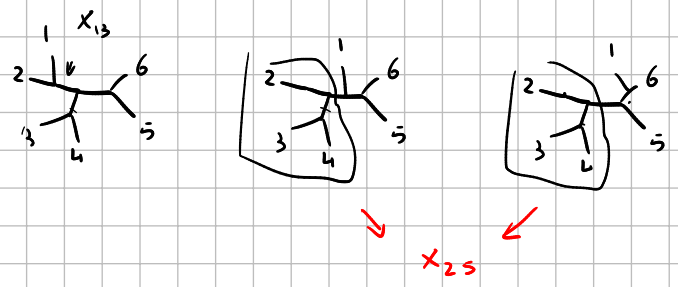
$$= a_1 (X_{15} - X_{13}) + a_2 X_{13} - a_3 X_{15} = 0$$

$$= \underline{X_{15}(a_1 - a_3) + X_{13}(a_2 - a_1)}$$

Since X_{13}, X_{15} are arbitrary it must be $a_1 = a_2$ and $a_1 = a_3$

so $a_1 = a_2 = a_3$.

<u>alternative</u>



X_{2j} are arbitrary and X_{1j} fixed by X_{2j}

$$\begin{cases} X_{26} = -X_{13} \\ X_{2j} = X_{1j} - X_{13} \end{cases}$$

$X_{1j} = X_{2j} + X_{26}$

since X_{25} arbitrary

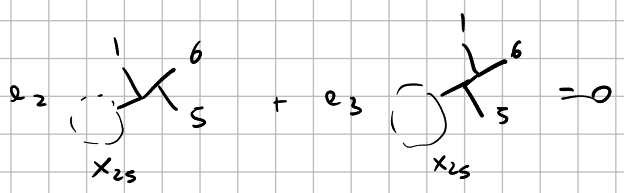
$$S_1 = (\text{--})_1 + \frac{1}{X_{25}} (\text{--})_2 = 0 \quad \Rightarrow \quad \begin{array}{l} (\text{--})_1 = 0 \\ (\text{--})_2 = 0 \end{array}$$

indeed:

eq 22 of Redine

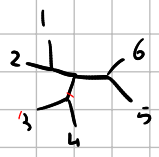
$$\begin{cases} a + b = 0 \\ a + 2b = 0 \end{cases} \Rightarrow \{$$

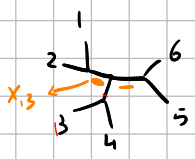
using that



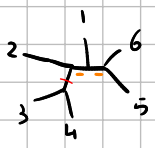
$$\frac{1}{X_{25}} \left(\frac{a_2}{X_{15}} + \frac{a_3}{X_{26}} \right) = 0 \qquad X_{26} = X_{16} - X_{13}$$

looks like eq. 22 is wrong (which is good!)

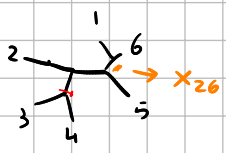
(otherwise the coefficient of  would be zero)



$$\frac{1}{x_{13}\,x_{15}}$$



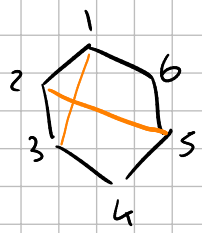
$$\frac{1}{x_{23}\,x_{15}}$$



$$\frac{1}{x_{25}\,x_{26}}$$

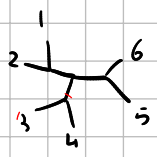
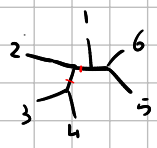
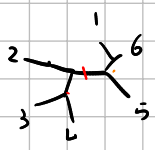
$x_{26} = -x_{13}$

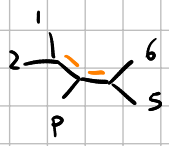
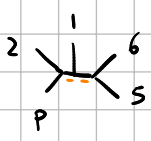
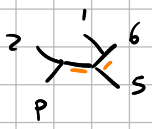
$$-\frac{1}{x_{25}\,x_{13}}$$



intersecting chords, not a triangulation

this does not correspond to a Feynman diagram

$S_1 =$ a_1  $+$ a_2  $+$ a_3 

$= \frac{1}{x_{35}}\Big($ a_1  $+$ a_2  $+$ a_3  $\Big)$

all graphs with $n=5$ that have 2 boundary propagator